

Course Syllabus

Statistical Methods for Data Analysis Course Syllabus

Course Information

STAT 5101 (4 cr.) | Spring 2026 | Online - Asynchronous

Course Prerequisites: Operationally none, a previous course in statistics will help at the beginning of the course

Course Description: In this course, you will understand variation and its consequences for drawing conclusions from data. You will become familiar with some standard statistical methods, when and how to use them, and how to interpret statistical results. You will be able to apply statistical principles to novel problems, as well as understand and employ the statistical language when communicating with other researchers. This class emphasizes the appropriate analysis of experimental data.

Course Personnel Information

Instructor

- **Name:** Ms. Majd Alsman
- **Email:** alsman@iastate.edu (<mailto:%20alsman@iastate.edu>)
- **Role:** I will be the instructor of the course and the main contact you should have if you encounter any issues or have questions. My role is to help you navigate through the course material and answer any questions via email or during office hours.
- I will post weekly [announcements](#) to help you stay on track with the course material and announcements about the Exams so keep an eye out for those announcements
- **Student Help Hours:** Monday and Wednesday, 1-2 pm CT in Teams using this [link](https://teams.live.com/join/9376119041920?p=6NILneTz4UDam9ah09)  (<https://teams.live.com/join/9376119041920?p=6NILneTz4UDam9ah09>). Or by appointment in Teams.
- **Contact:** alsman@iastate.edu

Instructor of record

- **Name:** Petruta Caragea
- **Role:** My role is to support the course leader, Ms. Alsman, throughout the semester. Please contact me if you encounter issues that cannot be resolved by Ms. Alsman and I will work with you to address them.

Preferred Contact: pcaragea@iastate.edu (<mailto:pcaragea@iastate.edu>)

Office Address: 1121C Snedecor Hall, phone: 515-294-5582

Course Content

STAT 5101 is divided into the 7 course modules below. Access to each successive module will be available as indicated on the schedule below. **Each module includes at least one lab and one homework assignment.**

- **Module 0** (<https://canvas.iastate.edu/courses/128073/modules/982252>) - Review of basic statistical concepts (optional module)
- **Module 1** (<https://canvas.iastate.edu/courses/128073/modules/982253>) - Basic principles, randomization and randomization tests (Week 1)
- **Module 2** (<https://canvas.iastate.edu/courses/128073/modules/982254>) - Model-based inference for comparisons of two sample means or proportions (Weeks 2 and 3)
- **Module 3** (<https://canvas.iastate.edu/courses/128073/modules/982255>) - Assumptions, diagnostic tools, consequences if violated (Weeks 3 and 4)
- **Module 4** (<https://canvas.iastate.edu/courses/128073/modules/982257>) - Comparing more than 2 groups (ANOVA) and “After the F test” methods (Weeks 5 and 6)
- **Module 5** (<https://canvas.iastate.edu/courses/128073/modules/982258>) - Introduction to Regression (Simple Linear and Logistic) (Weeks 7 and 8)
- **Module 6** (<https://canvas.iastate.edu/courses/128073/modules/982260>) - Multiple Regression (Weeks 9, 10 and 11)
- **Module 7** (<https://canvas.iastate.edu/courses/128073/modules/982261>) - Introduction to factorial ANOVA and blocking (Week 12)

Course Learning Objectives (CLOs)

Upon completing this course, you will be able to do the following:

- **CLO 1: Summarize and visualize** data from experimental studies to reveal patterns and variation.
- **CLO 2: Apply parametric** methods (e.g., t-tests, ANOVA, regression) to analyze experimental data.
- **CLO 3: Apply non-parametric** methods (e.g., rank-based tests, distribution-free methods) when parametric assumptions are not appropriate.
- **CLO 4: Compare two groups** using appropriate statistical procedures and interpret the results.
- **CLO 5: Compare multiple groups** using analysis of variance and related approaches.
- **CLO 6: Model and interpret relationships** between a response and one or more continuous predictors using regression and related methods.
- **CLO 7: Evaluate how features of study design** (e.g., randomization, blocking, sample size, independence) affect the choice of statistical methods.
- **CLO 8: Assess the assumptions** underlying statistical procedures and use diagnostic tools to check whether they are met.
- **CLO 9: Interpret and communicate** results in context, recognizing the consequences of unmet assumptions.

Course Materials

All materials are available in Canvas. Here is a list of the typical materials provided:

- **Lecture Notes in PDF format:** many of these notes have spaces for you to take notes as you view the content videos.
- **Videos over Lecture Notes:** short videos containing presentations of the course materials.
- **Lab activities:** hands-on illustrations of statistical principles and computing practice for each module.
- **R Activities:** PDF format files explaining how to use R to build conceptual understanding of the course content. R scripts for practice are provided when appropriate, for practice purposes.
- **Optional but recommended:** While there is no required textbook for the course, I highly recommend you obtain access to the *Statistical Sleuth* (either edition 2 or 3), by Ramsey, F.L. and Schafer, D.W., 2012. This book is among the best statistics texts available for a course like Stat 5101. This book is organized in a non-traditional way because it emphasizes how statistical methods get used to answer scientific questions. Each chapter starts with two (or more) case studies that illustrate the use of methods about to be discussed. This is followed by sections that describe those methods and a section of related issues. Sometimes, however, these problems are more complicated than is optimal for introducing statistical concepts or methods. Our lectures will cover the same concepts, but we will often use different examples and may use a different presentation.

Course Technologies

- **Canvas** - learning management system. See the [Support and Resources \(https://canvas.iastate.edu/courses/128073/pages/g1e6889f8d4b5be9094f661a21fa786b8\)](https://canvas.iastate.edu/courses/128073/pages/g1e6889f8d4b5be9094f661a21fa786b8) page for more information.
- **Echo360** - video capture software for course recordings. Two screens were recorded for course videos, one capturing the instructor and the other the computer. You can choose to watch one or the other or both while viewing each video. Closed Captioning is available for all videos by selecting the CC icon on each video.
- **R** - statistical software package. In STAT 5101, we use R to fit models and conduct data analysis. You can find this [link \(https://canvas.iastate.edu/courses/128073/pages/introduction-to-r\)](https://canvas.iastate.edu/courses/128073/pages/introduction-to-r) if you are new to the software.
- **Teams** - for synchronous online interactions. Links to Teams rooms for help hours are available on the Home page in Canvas and in this syllabus.

Assessments

Homework Assignments

This course will include weekly practice homework assignments. These assignments are designed to help students review the main ideas from the lecture videos, labs, and course notes, and to prepare for the exams and course project.

Homework assignments will **not** be submitted for grading, except for Homework 1. Homework 1 will be completed through Canvas and will ask for general course information, such as confirming access to course materials, and providing basic information needed for course planning.

For the remaining homework assignments, students are expected to work through the problems on their own as weekly practice. These assignments are an important part of exam preparation, even though they will not receive a numerical grade or be submitted to Canvas.

A recorded video reviewing the main points from the homework assignment, main points covered in the modules, will be posted in Canvas. These videos will focus on the most important concepts, common mistakes, problem-solving strategies, and connections to the exams. Students should use these videos to check their understanding and identify areas where they need more practice.

The purpose of the homework assignments is to give students regular practice in statistical reasoning, interpretation, calculations, and applied problem-solving. Students are encouraged to complete the homework before watching the weekly homework review video so that the video can be used as a self-check.

Homework solutions or selected solution explanations may be provided when appropriate. Students are responsible for reviewing the weekly homework materials and homework review videos as part of their exam preparation.

Since homework assignments after Homework 1 are not submitted for grading, there is no late homework submission policy for these assignments. Exam preparation and success in the course will depend strongly on keeping up with the weekly practice materials.

Exams

There will be three exams in this course, all hosted through the Testing Center. If you have questions regarding their policies or requirements, reach out to the Testing Center team by emailing testcenterhelp@iastate.edu (<mailto:testcenterhelp@iastate.edu>)

In-Person Proctoring and In-Person, Off-Campus Proctoring (Canvas Exam)

Students either residing in the Ames area or taking an in-person class at Iowa State must register for in-person proctoring at one of the physical testing center locations. Use the RegisterBlast link in the course navigation menu to schedule your exams or visit the testing center's website directly. If you don't have an Ames address but still want an in-person proctor (or if you require additional accommodations outside of what the testing center can provide), then you are responsible and required to arrange for an authorized external proctor to supervise your exams in this course. See the Testing Center's official guidelines below or visit their [Syllabus Statements \(https://www.testcenter.iastate.edu/faculty/syllabus-statements/\)](https://www.testcenter.iastate.edu/faculty/syllabus-statements/) page

Official Testing Center Statement:

The [Testing Center \(https://www.testcenter.iastate.edu/\)](https://www.testcenter.iastate.edu/) will host exams in this course. For questions, reach out to the Testing Center team by emailing testcenterhelp@iastate.edu (<mailto:testcenterhelp@iastate.edu>).

For students within the Ames area: [Schedule your required exam appointment \(https://www.testcenter.iastate.edu/students/schedule-your-exam/\)](https://www.testcenter.iastate.edu/students/schedule-your-exam/) in advance and adhere to the guidelines for scheduling, rescheduling, and canceling. Take your exam at any open Testing Center location. Visit the Testing Center's website for additional information on [what to expect \(https://www.testcenter.iastate.edu/students/4229-2/\)](https://www.testcenter.iastate.edu/students/4229-2/) when using their facilities. The Testing Center will have modified procedures during [Finals Week \(https://www.testcenter.iastate.edu/finals-week/\)](https://www.testcenter.iastate.edu/finals-week/), so please plan accordingly.

For students outside of the City of Ames: You are responsible for and required to arrange for an authorized external proctor to supervise your exams in this course. Please choose a proctor who meets the [Testing Center's Off-Campus Proctoring policies](https://www.testcenter.iastate.edu/students/proctoring-for-online-students/) (<https://www.testcenter.iastate.edu/students/proctoring-for-online-students/>) and follow instructions for selecting and submitting proctors. Upon approval, your proctor will receive exam instructions and will be responsible for supervising your exam session.

Virtual Proctoring (Canvas Exam)

Virtual proctoring is only available for students residing outside the city of Ames with no in-person classes or obligations at Iowa State. Virtual proctoring is provided by a third-party vendor: SmarterProctoring. Primarily, they use an automated method of proctoring, and you can schedule your exam appointments by clicking on the SmarterProctoring link in the course navigation menu. If you prefer an in-person, off-campus proctor, an opt-out option is available, but it will be the responsibility of the student to source a proctor that meets the Testing Center's requirements. See the Testing Center's official guidelines below or visit their [Syllabus Statements](https://www.testcenter.iastate.edu/faculty/syllabus-statements/) (<https://www.testcenter.iastate.edu/faculty/syllabus-statements/>) page.

Official Testing Center Statement:

The [Testing Center](https://www.testcenter.iastate.edu/) (<https://www.testcenter.iastate.edu/>) will host exams in this course. For questions, reach out to the Testing Center team by emailing testcenterhelp@iastate.edu (<mailto:testcenterhelp@iastate.edu>).

For students within the City of Ames: [Schedule your required exam appointment](https://www.testcenter.iastate.edu/students/schedule-your-exam/) (<https://www.testcenter.iastate.edu/students/schedule-your-exam/>) in advance and adhere to the guidelines for scheduling, rescheduling, and canceling. Visit the Testing Center's website for additional information on [what to expect](https://www.testcenter.iastate.edu/students/4229-2/) (<https://www.testcenter.iastate.edu/students/4229-2/>) when using their facilities. The Testing Center will have modified procedures during [Finals Week](https://www.testcenter.iastate.edu/finals-week/) (<https://www.testcenter.iastate.edu/finals-week/>), so please plan accordingly.

For students outside the City of Ames: [Virtual Proctoring is provided by SmarterProctoring](https://iastate.box.com/s/xyvb3veszj5pkme2c6o991fo8doimet1)  (<https://iastate.box.com/s/xyvb3veszj5pkme2c6o991fo8doimet1>). An opt-out option is provided.

After each exam, selected review materials or solution explanations may be posted to help students understand common mistakes and prepare for later course material. Students are expected to review these materials because later modules build on earlier concepts.

Group Projects

All students will complete **one team project** (assigned by the course instructor) during the semester. Detailed project guidelines will be made available in Canvas and should be studied and followed carefully. There will be two milestone assignments for the project due during the semester, leading up to the submission of the final project.

Group Member Evaluation

Collaboration is an important skill and requires practice and dedication to make it work well for the team. Your future career promotions will depend on whether you have contributed fairly to your group's work. In the spirit of keeping each group member accountable for their effort, there will be a peer review form assigned at the end of the semester. Instructions for completing the peer evaluation form will be provided when it is released. Every

student will receive a score from their teammates, which is based on their assessment of each member's contribution to the homework assignments and the team project.

The **final project grade** will be the sum of the grades on each of the individual components (two milestones, final project presentation and peer review)

Grading

Grade Distribution

Your grade in this course includes the following participation areas:

Course Components	Total Percentage	Timing
Exam 1	20%	Week 5
Exam 2	20%	Week 8
Exam 3	20%	Week 12
Group project	40%	2 milestone reports and the final report due Week 12

Most Strict Grading Scheme

Below is the most strict grading scheme that will be used for semester grades in STAT 5101. It is possible (and very likely) this scheme will be relaxed somewhat before the end of the semester based on performance on homework, exams and group project. Given the asynchronous nature of the course, the determination of the final grading scheme will not take place until the end of the semester.

Grade	Range
A	93% to 100%

Grade	Range
A-	90% to < 93%
B+	87% to < 90%
B	83% to < 87%
B-	80% to < 83%
C+	77% to < 80%
C	73% to < 77%
C-	70% to < 73%
D+	67% to < 70%
D	63% to < 67%
D-	60% to < 63%
F	0% to < 60%

Course Policies regarding the use of AI

Students may use artificial intelligence tools in this course to support learning outside of exams. Examples include brainstorming project ideas, reviewing concepts, improving clarity in writing, organizing explanations, or checking grammar.

AI tools should support your learning, not replace your thinking. Students are responsible for understanding, verifying, and explaining any work they submit. AI tools sometimes produce incorrect or misleading answers, so students must check all AI-assisted work carefully.

AI tools are not allowed during exams. All three exams are proctored and completed through Canvas. During exam periods, students may only use the materials and resources explicitly allowed in the exam instructions. Using AI tools during an exam is considered academic misconduct.

For projects or any submitted written work where AI tools are used, students must include a short AI-use statement at the end of the submission. The statement should describe how AI was used. The statement should

distinguish between help with wording, grammar, or organization and help that contributed ideas, analysis, calculations, code, or other intellectual content.

Failure to acknowledge AI-generated or AI-assisted content in submitted work may be treated as academic misconduct under university policy.

Examples of acceptable AI-use statements will be provided in the Start Here module in [Canvas \(https://canvas.iastate.edu/courses/128073/pages/responsible-use-of-ai-tools\)](https://canvas.iastate.edu/courses/128073/pages/responsible-use-of-ai-tools).

ISU Required Syllabus Statement

Free Expression

Iowa State University supports and upholds the First Amendment protection of [freedom of speech \(https://freespeech.iastate.edu/\)](https://freespeech.iastate.edu/) and the principle of [academic freedom](https://www.iowaregents.edu/plans-and-policies/board-policy-manual/310-academic-freedom)  [\(https://www.iowaregents.edu/plans-and-policies/board-policy-manual/310-academic-freedom\)](https://www.iowaregents.edu/plans-and-policies/board-policy-manual/310-academic-freedom) in order to foster a learning environment where open inquiry and the vigorous debate of a diversity of ideas are encouraged. Students will not be penalized for the content or viewpoints of their speech as long as student expression in a class context is germane to the subject matter of the class and conveyed in an appropriate manner.

In effect Summer 2024: No employee, student, applicant, or campus visitor is compelled to disclose their pronouns. Anyone may voluntarily disclose their own pronouns.

ISU Recommended Syllabus Statement

Academic Dishonesty

The class will follow Iowa State University's policy on academic misconduct ([5.1 in the Student Code of Conduct \(https://www.policy.iastate.edu/policy/SDR\)](https://www.policy.iastate.edu/policy/SDR)). Students are responsible for adhering to university policy and the expectations in the course syllabus and on coursework and exams, and for following directions given by faculty, instructors, and Testing Center regulations related to coursework, assessments, and exams. Anyone suspected of academic misconduct will be reported to the [Office of Student Conduct in the Dean of Students Office \(https://www.studentconduct.dso.iastate.edu/academic-misconduct/armfacultystaff\)](https://www.studentconduct.dso.iastate.edu/academic-misconduct/armfacultystaff). Information about academic integrity and the value of completing academic work honestly can be found in the [Iowa State University Academic Integrity Tutorial](https://iastate.pressbooks.pub/academicintegrity/)  [\(https://iastate.pressbooks.pub/academicintegrity/\)](https://iastate.pressbooks.pub/academicintegrity/).

Accessibility Statement

Iowa State University is committed to advancing equity, access, and inclusion for students with disabilities. Promoting these values entails providing reasonable accommodations where barriers exist to students' full participation in higher education. Students in need of accommodations or who experience accessibility-related barriers to learning should work with

Student Accessibility Services (SAS) to identify resources and support available to them. Staff at SAS collaborate with students and campus partners to coordinate accommodations and to further the academic excellence of students with disabilities. Information about SAS is available online at [www.sas.dso.iastate.edu \(https://sas.dso.iastate.edu/\)](https://sas.dso.iastate.edu/), by email at [accessibility@iastate.edu \(mailto:accessibility@iastate.edu\)](mailto:accessibility@iastate.edu), or by phone at 515-294-7220.

Discrimination and Harassment

Iowa State University does not discriminate on the basis of race, color, age, ethnicity, religion, national origin, pregnancy, sexual orientation, gender identity, genetic information, sex, marital status, disability, or status as a U.S. Veteran. Inquiries regarding non-discrimination policies may be directed to Office of Equal Opportunity, 3410 Beardshear Hall, 515 Morrill Road, Ames, Iowa 50011, Tel. 515-294-7612, Hotline 515-294-1222, email [eooffice@iastate.edu \(mailto:eooffice@mail.iastate.edu\)](mailto:eooffice@iastate.edu)

Mental Health and Wellbeing Resources

As a Cyclone, you can access 24/7 resources and people dedicated to helping you achieve your goals and be your best in and out of the classroom. Whether you need academic support or just someone to talk to, we're here for you at Cyclone Support ([cyclonesupport.iastate.edu \(https://cyclonesupport.iastate.edu/\)](https://cyclonesupport.iastate.edu/)).

Prep Week

This class follows the Iowa State University Prep Week policy as noted in section 10.6.4 of the [Faculty Handbook \(https://www.provost.iastate.edu/policies/faculty-handbook\)](https://www.provost.iastate.edu/policies/faculty-handbook).

Religious Accommodation

Iowa State University welcomes diversity of religious beliefs and practices, recognizing the contributions differing experiences and viewpoints can bring to the community. There may be times when an academic requirement conflicts with religious observances and practices. If that happens, students may request a reasonable accommodation for religious practices. In all cases, you must put your request in writing. The instructor will review the situation in an effort to provide a reasonable accommodation when possible to do so without fundamentally altering a course. For students, you should first discuss the conflict and your requested accommodation with your professor at the earliest possible time. You or your instructor may also seek assistance from the [Dean of Students Office \(https://www.studentassistance.dso.iastate.edu/\)](https://www.studentassistance.dso.iastate.edu/) at 515-294-1020 or the [Office of Equal Opportunity \(https://www.eoc.iastate.edu/\)](https://www.eoc.iastate.edu/) at 515-294-7612.

Student Code of Conduct

Students in this course are responsible for being familiar with the Iowa State University student rules and policies in the [Student Code of Conduct \(https://www.policy.iastate.edu/policy/SDR\)](https://www.policy.iastate.edu/policy/SDR) in the Policy Library.

Disclaimer: In extenuating circumstances, the schedule, policies, or procedures are subject to change. All efforts will be made not to alter assignment due dates or scheduled exams. Any modifications will be communicated in writing, verbally in class, and/or published in the Canvas course.